Silent but deadly? Physics says seldom, memory says mostly

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Abstract

The belief that a silent passage of flatus smells worse than an audible one appears in many languages but has not been tested. We follow a passage from its chemistry through sound and dispersion to a bystander's memory, combining measured odorant data, a lip-reed model of the anal valve, analytic and computational models of dispersion, and models of what an observer notices.

Methanethiol carries 43–94% of the summed odour activity of the three sulfur odorants, never less than hydrogen sulfide. Composition, not temperature, sets whether a passage rises or sinks, and gas-phase oxidation removes at most 0.4% of any odorant within two minutes in an office, so the odorants travel unchanged with the air. In the lip-reed model, 85% of releases in the measured range of passage volumes never set the valve oscillating; they are physically silent rather than quiet.

We prove that while transport is linear in concentration, a quieter release of the same gas cannot produce a higher peak concentration. Near the source, a loud jet is diluted before it slows to the speed of the surrounding air, by a factor proportional to the ratio of exit to ambient velocity, which we call the Silencer number. At equal mass this pre-dilution lets a silent release reach a neighbour's nose at a median 1.45 times the concentration of an audible one, often below the just-noticeable ratio of 1.28.

Silent passages are also smaller, and as emitted a silent passage is the stronger one at the neighbour in 48% of pairs.

In computational fluid dynamics of real scenes, the warm air rising along the body carries a release to different noses depending on its momentum. On a sofa the silent release rises behind the releaser's back to the ceiling and reaches neither nose within 30 s, while the audible one reaches both. In an elevator the silent release is stronger at every nose when a person stands behind the releaser and weaker at every nose when a wall does. Across these scenes a silent passage as emitted is the stronger one at a bystander's nose in 0–70% of pairs (mean 21%).

A silent passage can be noticed only by smell, so every silent passage a bystander remembers was smelled, while audible passages are remembered whether or not they were. With literature models of untrained observers, a remembered silent passage seems the stronger one in 68–93% of comparisons, against 32–51% over all passages. The releaser, who notices every release, is not subject to this selection.

1 Introduction

In the film *3 Idiots*,[1] a student nicknamed "Silencer" produces silent but foul flatulence and, in one scene, declares the silent release the deadliest. The belief behind the joke is old. An 1811 slang dictionary defines a related term as an escape "more obvious to the nose than ears",[2] and English has called such releases "silent but deadly" since at least 1970. A documentary survey of 32 languages (Section S7) finds the distinction between silent and loud releases lexicalised in at least five language families. The claim that the silent release smells worse is stated explicitly in Chinese and English sayings, and it is built into the definitions of smell-specific "silent" words that arose independently at least four times within Indo-European.

We found no test of the claim. The odour of flatus is well characterised: hydrogen sulfide, methanethiol and dimethyl sulfide carry it,[3,4] and the volume of a passage does not

predict its sulfur content.[5] Its sound has been attributed to vibration of the anal skin, like the buzzing lips of a brass player,[6] but no peer-reviewed study has measured it. The only data relating sound to smell are informal and not peer reviewed: in 1 619 recorded passages, louder ones were very slightly smellier by a gas-sensor reading ($R^2 = 0.016$), and none was silent.[7] Their direction agrees with what we find, but they cannot test a claim about silent passages and human noses.

The question is narrow, but answering it needs three results that are not specific to flatus.

- *Where does a release next to a body go?* The warm convective boundary layer around a person rises along the body to the breathing zone[8,9] and carries pollutants released near the body with it.[10] Near-body sources therefore give far higher exposure per unit mass than sources elsewhere in the room,[11,12] and CFD can follow particles shed from skin and clothing in this flow.[13] These studies concern passive sources and the occupant's own exposure. We show that the momentum of a release decides whether the boundary layer takes it, and so which of several nearby noses it reaches.

- *Can a slower release give a higher peak?* We prove that under linear transport it cannot, in any room, posture or flow. A slower release can win only in the non-linear near field, by a factor bounded by a velocity ratio.

- *How can a sensory belief arise that the physics does not support?* When an event can be noticed through either of two senses, selecting the noticed events makes the two senses appear related (Berkson's bias[14,15]). We show that this selection alone can explain the belief in a population where it is false, and we derive predictions that can test this explanation.

Both halves of the claim are also ambiguous. A release can be silent because it never sets the anal valve oscillating, or because background noise masks it. "Deadlier" can mean a higher peak at a nose, a larger total dose, or a stronger memory. The comparison can be between the same gas released in two ways (equal mass), between passages as they are

actually emitted, or between the passages an observer happens to notice. These bases give different answers, and we keep them apart throughout (Section 2.1).

We follow one passage from its chemistry through sound and dispersion to memory (Figure S1). Chemistry fixes which odorant carries the smell, what sets the buoyancy of a passage, and whether the odorants survive the trip to a nose. A lip-reed model of the valve gives the share of releases that are physically silent. Two theorems bound what linear transport can do to the silent/audible ratio; an analytic model of starting jets and puffs, checked against canonical jets and body plumes, gives the ratio over the range of releases; and computational fluid dynamics (CFD) of two people on a sofa, four in an elevator and two standing face to face shows where the body's own airflow takes each release. A third theorem and literature models of observers give how often a remembered silent passage seems the stronger one.

2 Definitions and theory

2.1 What "deadlier" means

A passage is a release of volume V over a duration T through an effective orifice of diameter d, with exit velocity $u_0 = 4V/(\pi d^2 T)$. We take its *deadliness* for a given person to be the perceived intensity of the worst single inhalation,

$$I = k\,\mathrm{OAV}_\mathrm{b}^{n}, \qquad \mathrm{OAV}_\mathrm{b} = \max_{\text{breaths}} \frac{\int_\mathrm{insp} \dot{V}_\mathrm{insp}(t)\,\mathrm{OAV}(t)\,\mathrm{d}t}{\int_\mathrm{insp} \dot{V}_\mathrm{insp}(t)\,\mathrm{d}t}, \tag{1}$$

where $\mathrm{OAV} = \sum_i c_i/c_{\mathrm{th},i}$ is the summed odour activity value of hydrogen sulfide, methanethiol and dimethyl sulfide at the nose and n is the Stevens exponent.

The breath is the natural unit for three reasons.

- Perceived intensity saturates within a single natural sniff: one sniff is rated like seven, and the optimum lies between 0.4 and 0.6 s.[16,17]
- Within that window intensity still integrates: longer H_2S pulses of 100–300 ms are

rated stronger.[18]

- The sniff is the unit of olfactory processing,[19,20] and regulatory odour-impact assessment also works with per-breath peaks rather than hourly means.[21]

Cumulative exposure is reported as a secondary metric.

Every comparison below states its basis:

(A) *equal mass*: the same gas, released quietly or loudly;

(B) *as emitted*: silent and audible passages drawn from their own volume distributions;

(C) *as remembered*: only the passages an observer noticed.

Folk experience is basis (C). Physics answers (A) and (B).

2.2 Linear transport bounds the silent advantage

Odorants at sub-ppm levels are a passive scalar: the air carries them without being affected by them. Once the release has lost its own momentum and buoyancy, the concentration at a fixed nose is a linear functional of the release history $q(t)$ (volume of flatus per unit time):

$$C(t) = \int G(t-s)\, q(s)\, \mathrm{d}s = (G * q)(t), \qquad \int q\, \mathrm{d}s = V, \tag{2}$$

where the Green's function G encodes everything about the room, the bodies, the furniture and the ventilation. Time invariance of G requires the flow to be statistically stationary over the release, which holds for the quasi-steady convective boundary layers and room flows considered here.

Theorem 1. Let a silent release be a smoothed version of an audible one, $q_\mathrm{s} = q_\mathrm{a} * \kappa$ with $\kappa \geq 0$ and $\int \kappa = 1$. Then, for any G:

(i) the time-integrated dose $\int C\, \mathrm{d}t = V \int G$ is the same for both;

(ii) the peak concentration of the silent release does not exceed that of the audible one.

Proof. (i) follows from integrating Eq. (2) over t. For (ii), $C_{\mathrm{s}} = G * q_{\mathrm{a}} * \kappa = C_{\mathrm{a}} * \kappa$, and Young's inequality gives $\|C_{\mathrm{s}}\|_\infty \le \|C_{\mathrm{a}}\|_\infty \, \|\kappa\|_1 = \|C_{\mathrm{a}}\|_\infty$. □

A top-hat release lasting nT is a top-hat lasting T convolved with $\kappa = n^{-1} \sum_{j=0}^{n-1} \delta(t - jT)$, so stretching a release by an integer factor is covered. The same argument applies to the breath-averaged OAV of Eq. (1), which is itself a convolution of C with a non-negative kernel. In the linear regime, therefore, a quiet release can never be deadlier than a loud release of the same gas, whatever the posture, room or flow.

2.3 Release mode is forgotten

Theorem 2. In a closed, mixed room of fluid volume V_{room}, $G(t) \to 1/V_{\mathrm{room}}$ as $t \to \infty$. Both release modes therefore approach $C = V/V_{\mathrm{room}}$, and their ratio tends to one. More generally, any difference between release modes is carried by the early, near-field part of G, and it decays with travel time and distance.

2.4 Where the theorems fail: momentum and buoyancy

Equation (2) fails only near the source, where the release still carries its own momentum and buoyancy and so changes the flow that carries it. Two effects survive this.

Near-field pre-dilution. A momentum-dominated release entrains ambient air until its self-induced velocity has decayed to the ambient velocity u_h (the body's convective boundary layer or room air). For a starting jet, a puff or a steady jet alike, the dilution at that hand-off is

$$D_1 \approx \beta \, \frac{u_0}{u_h} \equiv \beta \, \mathrm{Si}, \qquad \beta = O(1), \tag{3}$$

because entrained mass is inversely proportional to the remaining velocity (Section 3; jet and puff coefficients from refs. 22,23). We call $\mathrm{Si} = u_0/u_h$ the *Silencer number*. A audible release (in the analytic model $\mathrm{Si} > 280$) arrives pre-diluted; a silent one ($\mathrm{Si} \approx 5$) does not.

At equal mass, pre-dilution favours the silent release by at most the ratio of the two Silencer numbers.

Routing. Momentum also sets which flow the release joins, and different flows lead to different noses. This effect can favour either release. In the rooms of Section 4.4 it changes the ratio by more than pre-dilution does.

2.5 Selection biases what is remembered

Let H be the event that a passage is heard and S the event that it is smelled. An observer can only compare passages they noticed, $N = H \cup S$.

Theorem 3. Whatever the joint law of H and S,

$$P(S \mid \neg H, N) = 1, \qquad P(S \mid H, N) = P(S \mid H). \tag{4}$$

Proof. On $\neg H$, noticing requires smelling, so $\neg H \cap N = \neg H \cap S$. On H, N is certain, so conditioning on N adds nothing. □

Among remembered passages, then, every silent one stank, while audible ones stank only at the base rate p_S, even when sound and smell are independent. The gap $1 - P(S \mid H)$ is largest when smelly passages are rare. This is Berkson's bias:[14,15] selecting on an event that either of two causes can produce (here, noticing, produced by hearing or by smelling) makes the two causes appear related among the selected cases.

An observer whose noticing does not depend on smell has no such bias. The emitter, who feels every release, has N certain for all passages and so compares silent and audible releases at their true frequencies.

3 Methods

Every parameter, with its source, range and whether it is measured, derived or assumed, is listed in Table S4. Code and data are archived (Section S8).

3.1 Source

Passage volumes span 17–375 mL, with a median of 92 mL.[5] In the Monte Carlo analysis, volume V, duration T (0.3–20 s) and effective orifice diameter d (1–8 mm) are drawn log-uniformly. Neither has been measured for a passage. The upper bounds come from anorectal physiology: flatus passes during an anal relaxation lasting 19 ± 6 s,[24,25] and the mid anal canal widens to about 8 mm only at balloon pressures above those that drive flatus.[26–28] The lower bounds are assumed.

Sulfur gas concentrations are taken from 87 passages of 16 volunteers:[3] H_2S 1.06 µmol L^{-1}, methanethiol 0.21 µmol L^{-1} and dimethyl sulfide 0.08 µmol L^{-1}. They vary between passages by a log-normal factor with $\sigma_{\ln} = 1.1$, derived from the reported standard errors, drawn independently of volume because sulfur concentration does not correlate with passage volume.[5] Buoyancy is computed from measured compositions and is signed (Section 3.2); it is drawn independently of volume because, across the 16 subjects of ref. 5, it does not correlate with mean passage volume (Spearman -0.15).

3.2 Odorant chemistry

Odour activity. Each sulfur gas contributes an odour activity value (OAV), its mole fraction over its detection threshold, $\mathrm{OAV}_i = x_i/\theta_i$. The passage's OAV is their sum; we assume additivity and do not model mixture suppression. We compare two measured compositions:

- 87 passages of 16 volunteers,[3] used throughout the dispersion model;
- 28 passages measured by gas chromatography: H_2S 149, methanethiol 216 and dimethyl sulfide 144 nmol L^{-1}.[4]

Three threshold sets bracket the observer: a trained panel,[29] population geometric means[30,31] and the lowest values in the literature, as compiled in ref. 4.

Removal check. Zinc acetate precipitates H_2S and thiols; activated charcoal adsorbs all three gases. For both treatments, ref. 3 reports the gas concentrations and two judges' ratings before and after. We predict the rating ratio as $(\mathrm{OAV}_{\mathrm{after}}/\mathrm{OAV}_{\mathrm{before}})^n$ with $n = 0.20$–0.45,[32] without fitting.

Buoyancy. The reduced gravity $g' = g(\rho_a - \rho_g)/\rho_a$ of each of 16 measured compositions[5] is split into a thermal part, $g(1 - T_a/T_g)$, for gas at 37 °C in air at 22 °C, and a compositional part, $g(M_a - M_g)/M_a$, at equal temperature with air at 50% relative humidity. The water-vapour content of flatus has not been measured, so we bracket dry and saturated gas.

Gas-phase and surface losses. Gas-phase losses use IUPAC rate constants at 298 K for reaction with OH, O_3, NO_3 and Cl,[33] with oxidant levels measured in an office: OH 0.6–$9.8 \times 10^5\,\mathrm{cm^{-3}}$ and O_3 0.7–2.6 ppb.[34] Two assumed upper cases add OH $10^6\,\mathrm{cm^{-3}}$ with O_3 40 ppb, and Cl $10^6\,\mathrm{cm^{-3}}$ during hypochlorite cleaning. No deposition velocity v_d has been published for these gases on indoor surfaces, so surface loss is bracketed as $k = v_d\,S/V$ with $S/V = 3\,\mathrm{m^{-1}}$.

Dilution in a panel test. Ref. 4 reports the lowest source concentration that every panel member detected when 100 mL was released below the nose. Divided by a detection threshold, it gives the dilution between release and nose in that test, which we compare with the breath-averaged dilution of the analytic model.

3.3 Acoustics

The anal canal is modelled as a lip reed, a soft-tissue valve that self-oscillates in the flow through it, as the lips do on a brass instrument. It oscillates when the driving pressure $\frac{1}{2}\rho(u_0/C_\mathrm{d})^2$ exceeds a threshold $P_\mathrm{th} = 0.6$–2.5 kPa (nominal 1.2 kPa), with discharge coefficient $C_\mathrm{d} = 0.85$. The closest measured soft-tissue sphincter source, the pharyngo-oesophageal segment of oesophageal speech, sustains phonation down to 1.1–2.5 kPa,[35–37] which bounds

the onset from above; lips on a brass instrument start at about 0.6 kPa.[38] This gives an onset velocity $u_{\mathrm{onset}} \approx 42\,\mathrm{m\,s^{-1}}$ (21–71).

An oscillating valve radiates as a harmonic monopole with harmonics falling by 12 dB per octave. Its fundamental is taken as 160 Hz (60–300 Hz), between oesophageal speech at 60–150 Hz[35,39–41] and recordings of flatus near 270 Hz, which are not peer-reviewed. A release counts as *audible* when at least one harmonic exceeds both the ISO 226 hearing threshold[42] and the masked threshold of the background, computed from the critical ratio.[43] Backgrounds range from a 35 dB(A) living room to a 77 dB(A) subway car.

3.4 Analytic dispersion model

Dispersion is modelled in two phases.

1. *Near field.* The release behaves as a starting jet. Its tip follows $x = \Gamma(\dot{Q}_0 u_0)^{1/4} t^{1/2}$, and once the source stops it follows the puff law $x^4 = C_p^4 Q_0 u_0 t$, with $\Gamma = 2.8$ and $C_p = 2.6$.[22,23] The centreline dilution is $K_c d^*/x$ with $K_c = 5.11$. In parallel, a buoyant thermal grows with entrainment coefficient $\alpha = 0.25$,[44] and the larger of the two dilutions is kept. This phase ends when the self-induced velocity has fallen to the ambient velocity u_h (0.05–0.25 $\mathrm{m\,s^{-1}}$), which gives Eq. (3).

2. *Transport.* The cloud is then advected to the nose and spreads as a Gaussian puff, $\sigma^2 = \sigma_0^2 + 2Kt$, with indoor eddy diffusivity $K = 10^{-4}$–$1.3 \times 10^{-2}\,\mathrm{m^2\,s^{-1}}$.[45]

Two receptors are modelled: the releaser's own nose, reached by the convective boundary layer, the layer of warm air rising along the body,[8] at a speed taken from the CFD trajectories below; and a neighbour 0.8–1.4 m away, reached by room air. Concentrations are averaged over one inspiration.

We draw 20 000 pairs of silent and audible releases, each classified by the acoustic onset and evaluated with environmental parameters drawn from their ranges. Three simplifications

favour the silent release: turbulent-jet constants are used even for laminar silent releases, the nose sits on the cloud centreline, and the centreline concentration is used.

3.5 Computational fluid dynamics

We use OpenFOAM 13 with transient Reynolds-averaged Navier–Stokes equations (RANS), the k–ω SST turbulence model, the Boussinesq approximation for buoyancy and the energy equation in internal-energy form.

Validation (Section S1). A free round jet at Re = 6700 reproduces the canonical decay constants (velocity 5.0, scalar $K_c = 4.3$–4.6) with a grid uncertainty of at most 3.7%. The convective boundary layer of a seated manikin is 40–56% faster than thermal-manikin data[8] and follows their trend with room temperature. Both biases favour the silent release: excess jet spreading over-dilutes audible releases by about 15%, and the fast boundary layer shortens travel times to the releaser's own nose by about 30%, which the analytic model corrects.

Scenes (Section S2). Manikins built from geometric primitives, with skin at 33 °C, stand or sit in rooms at 23 °C:

- a sofa: two seated people 0.6 m apart in a 3.2 × 3.2 × 2.6 m living room, with the releaser upright, reclined by 4.5 ° or leaning forward by 10 °, over 30 s (upright also 120 s);
- an elevator: four standing people in a closed 630 kg car, fan off, over 60 s, with the releaser in the front or the back row;
- a standing pair: two people face to face, noses 1.0 m apart, in the same living room, in still air and with office ceiling ventilation at 2.9 air changes per hour (EN 16798-1 Category II[46]), over 60 s.

Each scene is first run without a release until the body plumes have developed. From that flow field, the same 92 mL is released behind the releaser's buttocks, as an inlet that matches the release's volume flow and momentum: silent, 2 s through 8 mm at $u_0 = 0.9\,\mathrm{m\,s^{-1}}$; audible,

0.5 s through 2.4 mm at 41 m s^{-1}. The odorant is a passive scalar, sampled 2.5 cm in front of each nose, over 5 cm discs at nose height and over sinusoidal breathing cycles.

3.6 Perception and observers

The nose samples only during inhalation: sinusoidal flow at 12 breaths per minute, inspiratory fraction 0.42.[47] Perceived intensity follows Stevens' power law, $I \propto \mathrm{OAV}^n$, with $n = 0.28$ measured for flatus H_2S.[32] Two passages are distinguishable when their concentrations differ by more than the olfactory Weber fraction of 0.28,[48] i.e. a just-noticeable ratio of 1.28.

The OAVs of the dispersion model use trained-panel thresholds.[29] For untrained observers:

- population thresholds are 10–23 times higher;[30,31]
- detection is a per-passage probability $\Phi(\log_2 \mathrm{OAV}/\sigma)$ with $\sigma = 2$ binary steps,[30] since one person's threshold varies over time about as much as thresholds vary across people;[49]
- a bystander who is not expecting an odour needs 26 times the threshold for a 50% response, with $\sigma = 3.5$ binary steps.[30,50]

Distraction reduces noticing[51] but not the peak rated intensity,[52] so intensity is computed from the attentive OAV. Two variants follow: clothing removes 22% of the sulfide gas,[53] and a sound can make a distracted bystander attentive for that passage.

3.7 Uncertainty analysis

Uncertainty is propagated through the analytic model; one CFD run of 30–60 s costs 1–3 h on four cores, too many for a sampling design. First-order Sobol indices (the share of the variance of $\log_{10} R$ explained by one input alone) are estimated by binning the Monte Carlo sample, with bootstrap 90% intervals. Because the silent and audible releases of a pair are

drawn by rejection, the inputs are dependent and the indices are descriptive. The regime map shows $\log_{10} R$ at equal mass against the audible release's Silencer number and the travel time.

Passages as emitted in the simulated scenes. Each CFD scene gives the doses of one release volume at equal mass. The OAV is linear in concentration, so a passage of volume V following the same flow path gives OAV $\times\, V/92\,\mathrm{mL}$. For each of the 20 000 Monte Carlo pairs we scale the silent and the audible CFD doses by the pair's two volumes, and report $P(R > 1)$ and the shares above and below the just-noticeable ratio per scene and nose. In the analytic model this scaling is exact ($R_{\text{as emitted}} = R_{\text{equal mass}}\, V_s/V_l$). Applied to the CFD it assumes that the flow path does not depend on the volume: close for silent releases, whose momentum is negligible against the body plume, and an approximation for audible ones.

4 Results

4.1 The odorants and the gas that carries them

Which molecule. Over the two measured compositions and three threshold sets, methanethiol carries 43–94% of the summed OAV and never less than H_2S (Figure 1a).

- With the composition of ref. 3, methanethiol and H_2S carry about equal shares (43–54% and 43–46% with trained-panel and population thresholds). Methanethiol is one fifth as concentrated, but its threshold is 5–6 times lower.
- With the composition of ref. 4, or with the lowest reported thresholds, methanethiol carries 60–94%.

This differs from the conclusion drawn in ref. 3, where H_2S is the main sulfur gas and the one whose concentration the judges' malodour ratings followed. Weighted by odour thresholds, the same measured concentrations give methanethiol at least an equal share. The two datasets disagree on the concentrations themselves: H_2S is five times methanethiol

in one and slightly below it in the other, a difference ref. 4 attributes to the bean-rich diet of the first. In that study's panel test, gas mixtures with methanethiol above its flatus threshold smelled like most passages.

Composition does not enter the silent/audible comparison: transport is linear and acts on all three gases alike, so every ratio is the same for any composition. Absolute OAVs are 11–38% lower with the second composition.

The intensity law with real flatus. Zinc acetate lowers the summed OAV by a factor of 14. Stevens' law then predicts rating ratios of 0.31–0.59, and the two judges gave 0.28 and 0.59.[3] Activated charcoal lowers the OAV about 260-fold and is rated far weaker than the power law predicts, as power laws fail near threshold. The silent/audible comparison uses supra-threshold ratios, where the law held.

What makes a passage rise or sink. Composition sets most of the buoyancy (Figure 1b).

- For the median passage the compositional part of the reduced gravity ($+1.25\,\mathrm{m\,s^{-2}}$ dry, +1.40 saturated) is about three times the thermal part (+0.47).
- H_2 and CH_4 make flatus light and CO_2 makes it heavy; the compositional part falls with CO_2 content ($r = -0.80$).
- 6–12% of passages sink even while hot. After cooling to room temperature, 25–38% are heavier than air.

The dispersion model uses these signed values of g'.

Losses on the way to the nose (Table 1).

- In an office, gas-phase oxidation removes at most 0.4% of any of the three gases in 120 s, and hypochlorite cleaning less than 4%.
- Surface deposition is unconstrained by data. It removes 0.7% in 120 s at $v_d = 0.002\,\mathrm{cm\,s^{-1}}$ and 51% at $0.2\,\mathrm{cm\,s^{-1}}$. Treating the odorants as passive scalars therefore holds against chemistry by two to three orders of magnitude, and against deposition only if $v_d \lesssim 0.01\,\mathrm{cm\,s^{-1}}$.

- Deposition would remove most from the silent release, which stays close to the body and the seat, and would lower its doses further in the scenes of Section 4.4.

How dilute a passage arrives. The panel test of ref. 4 implies a dilution of 0.6–13 $\times 10^4$ between release and nose, depending on the threshold set. The breath-averaged dilution of the analytic model at the neighbour has a median of 1.8×10^3 (10–90%: 2.5×10^2–2.9×10^4). The panel value is biased upwards, because it requires the least sensitive panellist to detect but divides by the thresholds of the most sensitive. The analytic model is at the concentrated end of the panel range, as intended for an upper bound, and within about an order of magnitude of it.

Composition and volume. Larger passages contain more fermentation gas and less nitrogen (per passage, $r = +0.56$ for CO_2 and -0.50 for N_2, $n = 87$), while the sulfur concentrations do not change with volume ($|r| \leq 0.22$, not significant); sulfur mass therefore rises with passage volume.[5] At high production rates almost all flatus is fermentation gas.[54] The fermentation-gas fraction rises with output under bean, fibre and lactulose challenges in every study that measured both ($r = +0.86$ to $+0.99$, computed from the published tables;[55–60] in one person from 26% at 55 mL to 79% at 720 mL per 4 h[61]), whereas on an ordinary diet larger outputs carry more nitrogen.[62] The density of the gas does not follow the amount: carbon dioxide and hydrogen rise together, and the mean molar mass, 22–32 g mol^{-1} across diets, is set by the substrate's CO_2:H_2 ratio (its correlation with output ranges from -0.37 to $+0.90$ across seven studies). Composition therefore cannot link audibility to the amount of gas, since it enters audibility only through density (Section 3.3).

Table 1: Gas-phase lifetime of the odorants and the fraction lost in 120 s. Office oxidant levels are measured:[34] low, OH 0.6×10^5 cm^{-3} and O_3 0.7 ppb; high, OH 9.8×10^5 cm^{-3} and O_3 2.6 ppb. The cleaning case assumes Cl 10^6 cm^{-3}. Rate constants from ref. 33.

scenario	H_2S	CH_3SH	DMS
office, low	40 d / 0.003%	5.7 d / 0.03%	28 d / 0.005%
office, high	2.5 d / 0.06%	8.6 h / 0.4%	1.8 d / 0.08%
hypochlorite cleaning	3.5 h / 0.9%	1.2 h / 2.8%	0.8 h / 4.0%

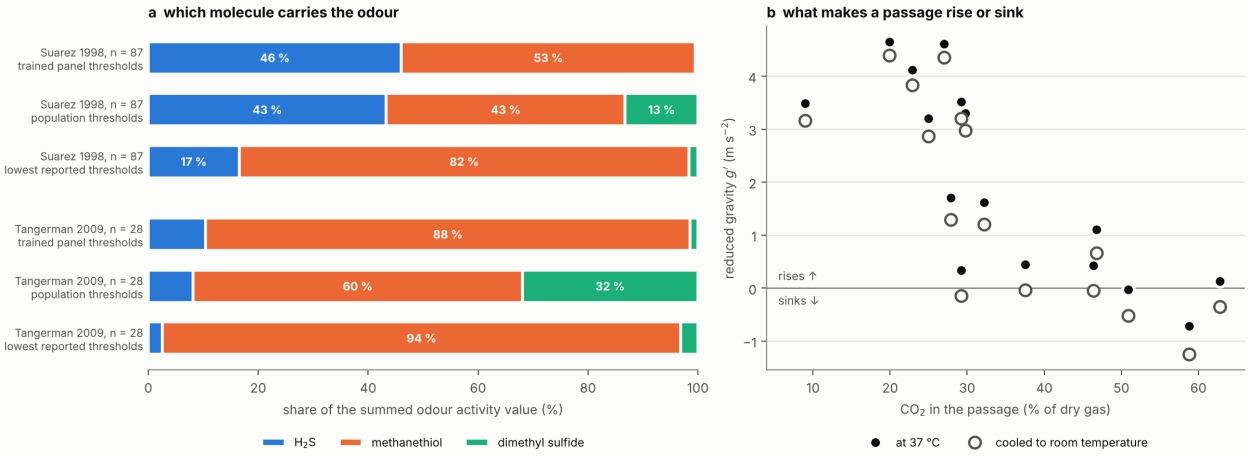


Figure 1: Odorants and buoyancy. (a) Share of the summed OAV carried by H_2S, methanethiol and dimethyl sulfide, for two measured compositions[3,4] under three threshold sets. (b) Reduced gravity of 16 measured compositions[5] against their CO_2 content, at 37 °C (dry basis; water saturation shifts these values by -0.04 to $+0.29\,\mathrm{m\,s^{-2}}$) and after cooling to room temperature.

4.2 Most releases are physically silent

In the lip-reed model, 85% of the releases sampled never set the valve oscillating (Figure 2); the other 15% are audible in any quiet room. No release in the sampled range oscillates without being heard in a quiet room: the smallest oscillating release, about $33\,\mathrm{mL\,s^{-1}}$ through a 1 mm orifice, gives a 47 dB fundamental at sofa distance (0.6 m), against an 18 dB hearing threshold at 160 Hz. A release that is not heard is therefore either physically silent or masked by background noise.

Masking needs loud backgrounds. In a restaurant or a subway car an oscillating release must exceed 0.36–$0.60\,\mathrm{L\,s^{-1}}$ at 0.6 m (1.8–$3.0\,\mathrm{L\,s^{-1}}$ at 3 m) to be heard, so only near-maximal passages are: 2–4% of all releases at a 160 Hz fundamental, none at 100 Hz and 6–13% at 250–300 Hz. Audible releases are fast, with a median exit velocity of $90\,\mathrm{m\,s^{-1}}$ through a 1.5 mm orifice.

4.3 The ratio in the analytic model

Table 2 gives the breath-averaged ratio $R = C_{\mathrm{silent}}/C_{\mathrm{audible}}$ for 20 000 random pairs.

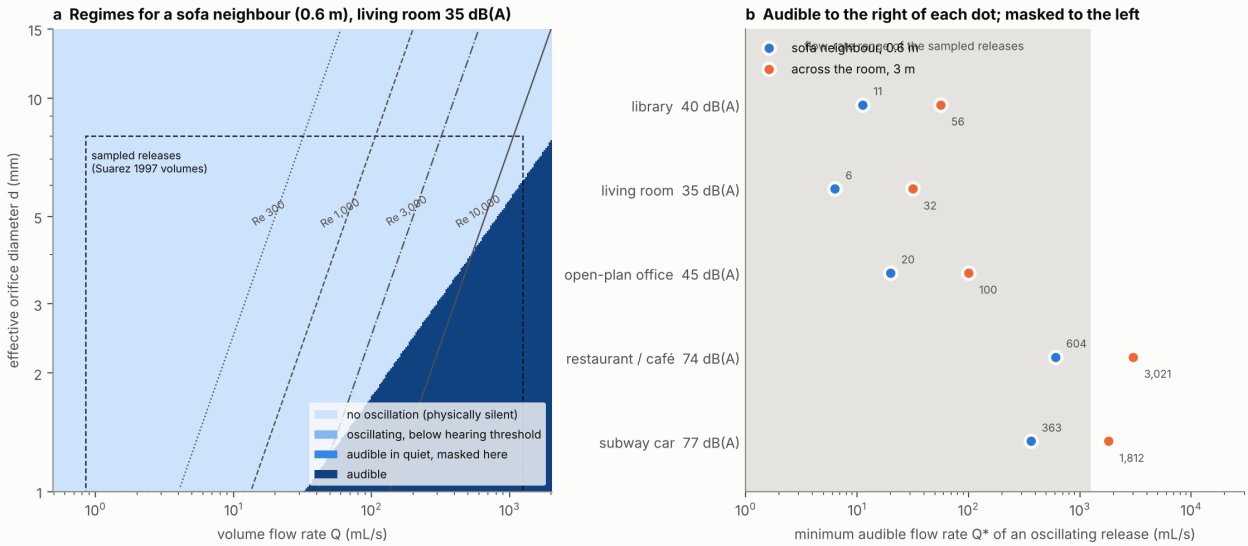


Figure 2: Acoustics. (a) Whether a release is heard, by volume flow rate and effective orifice diameter, for a neighbour at 0.6 m in a 35 dB(A) living room: the valve does not oscillate (physically silent); it oscillates but is masked here (audible in a quiet room); or it is audible. Dashed box: the sampled range of releases. Grey lines: jet Reynolds number. (b) Minimum audible flow rate of an oscillating release for five backgrounds, at 0.6 m and 3 m. The grey band is the flow-rate range of the sampled releases.

Table 2: Silent-to-audible concentration ratio in the analytic model (20 000 pairs).

receptor	basis	median R	5–95%	$P(R > 1)$
own nose	equal mass	1.94	0.94–16.4	0.93
own nose	as emitted	1.23	0.20–6.1	0.59
neighbour	equal mass	1.45	0.97–10.3	0.93
neighbour	as emitted	0.96	0.15–4.8	0.48
neighbour, denim	as emitted	0.62	0.11–3.3	0.31

Equal mass. At equal mass the audible release arrives pre-diluted by its own momentum (Eq. (3)), and the silent one is stronger at the neighbour in 93% of pairs. At the just-noticeable ratio of 1.28, the silent release is perceptibly stronger in 59% of pairs and indistinguishable in 39%.

As emitted. Audible passages are larger, and at the neighbour a silent passage is stronger in 48% of pairs: perceptibly stronger in 39%, perceptibly weaker in 42%. Trousers remove the jet momentum of both releases and lower $P(R > 1)$ to 0.31.

Which inputs matter. As emitted, the two passage volumes account for about 0.6 of the variance of $\log_{10} R$ at the neighbour; at equal mass, the eddy diffusivity K accounts for 0.37. Sphincter pressure, discharge coefficient, buoyancy and inhalation time each account for less than 0.5%: the valve parameters set whether a passage is heard, but hardly how strongly it smells.

Regime map (Figure 3a). The silent release is perceptibly stronger only at short travel times and high Silencer numbers of the audible release. At equal mass even a barely audible release is perceptibly weaker than a silent one, at the releaser's own nose (6.7 s; $R = 1.61$ at the acoustic onset) and at the neighbour (12.5 s; $R = 1.33$).

4.4 Real rooms: where the release goes

The analytic model carries every cloud straight to the nose. In the simulated rooms the warm air rising along the body takes the silent release with it, while the audible jet leaves it, and the two then reach different noses.

Sofa. The silent release rises in the gap between the releaser's back and the backrest, over the head and to the ceiling (Figure 4). Within 30 s its breathing-zone OAV stays at or below 10^{-10} at both noses. The audible jet hits the corner between seat and backrest, spreads sideways along the gap and reaches both breathing zones at OAV 0.2–0.6. With the releaser reclined by 4.5 ° or leaning forward by 10 °, the silent release again reaches neither breathing zone (at most 10^{-4}; Section S3). After about 70 s both clouds have returned from

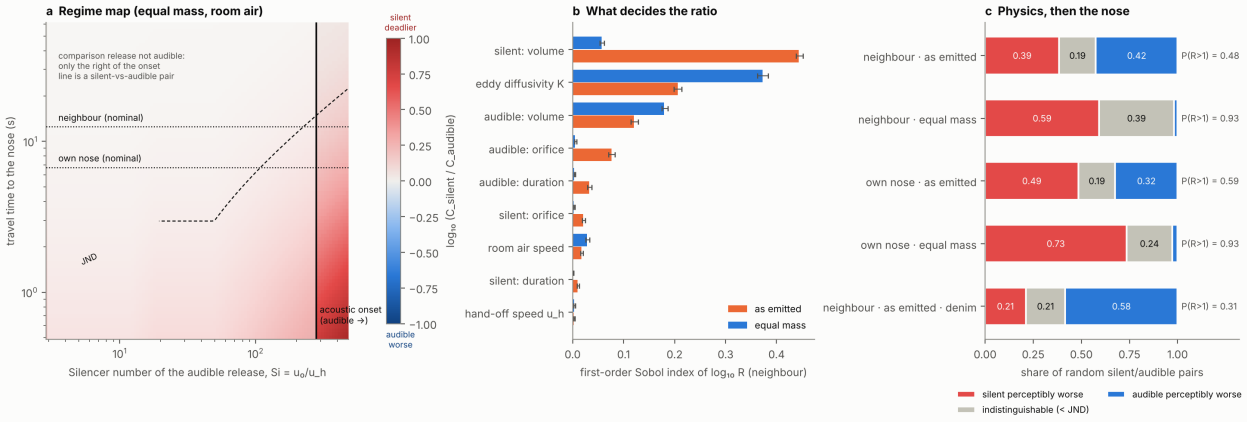


Figure 3: The ratio in the analytic model. (a) $\log_{10} R$ at equal mass and room-air mixing ($K = 10^{-3}\,\mathrm{m^2\,s^{-1}}$), over the Silencer number of the audible release and the travel time to the nose. The vertical line is the acoustic onset (Si = 279); to its left the comparison release is not audible (grey). The dashed contour is the just-noticeable ratio 1.28. Dotted lines mark the nominal travel times to the releaser's (6.7 s) and the neighbour's (12.5 s) nose. (b) First-order Sobol indices of $\log_{10} R$ at the neighbour, as emitted and at equal mass, with bootstrap 90% intervals. (c) Shares of random silent/audible pairs in which the silent release is perceptibly stronger, indistinguishable, or perceptibly weaker, with $P(R > 1)$.

the ceiling, and at the neighbour the two traces are indistinguishable (worst breath 0.39 for both), as Theorem 2 requires; at the releaser's own nose the mixed silent cloud is 1.5–2 times higher and touches the trained-panel threshold for about a second near 100 s.

Elevator. In a closed car with four standing occupants, the result depends on what stands behind the releaser. With a person behind (front row), the silent release rises in the 0.2 m gap into that person's breathing zone and is stronger at every nose ($R = 1.4$–4.3 at equal mass), reaching OAV 27 for the person behind. With the car wall behind (back row), it rises between back and wall, and the audible release is stronger at every nose ($R = 0.20$–0.55). With a front-row releaser the person behind reaches threshold first (8.8 s, against 17.6 s at the releaser's own nose), so the rule that whoever smells it first produced it holds for back-row releasers only. Onlookers who see only who reacts identify the releaser at 0.28–0.52, against 0.25 by chance (Section S4).

Standing pair. **[GAP: standing pair face to face, still air and office ventilation: results pending]**

Passages as emitted. Scaling each scene's equal-mass doses by the volumes of passages as emitted (Section 3.7) gives Figure 5. Silent passages are also smaller (median volume ratio 0.45), which lowers the silent advantage where the flow favours the silent release: for the person behind a front-row releaser, a silent passage is stronger in 35% of pairs. Over the nine bystander noses $P(R > 1)$ is 0–0.70, with an equal-weight mean of 0.21, against 0.48 in the analytic model.

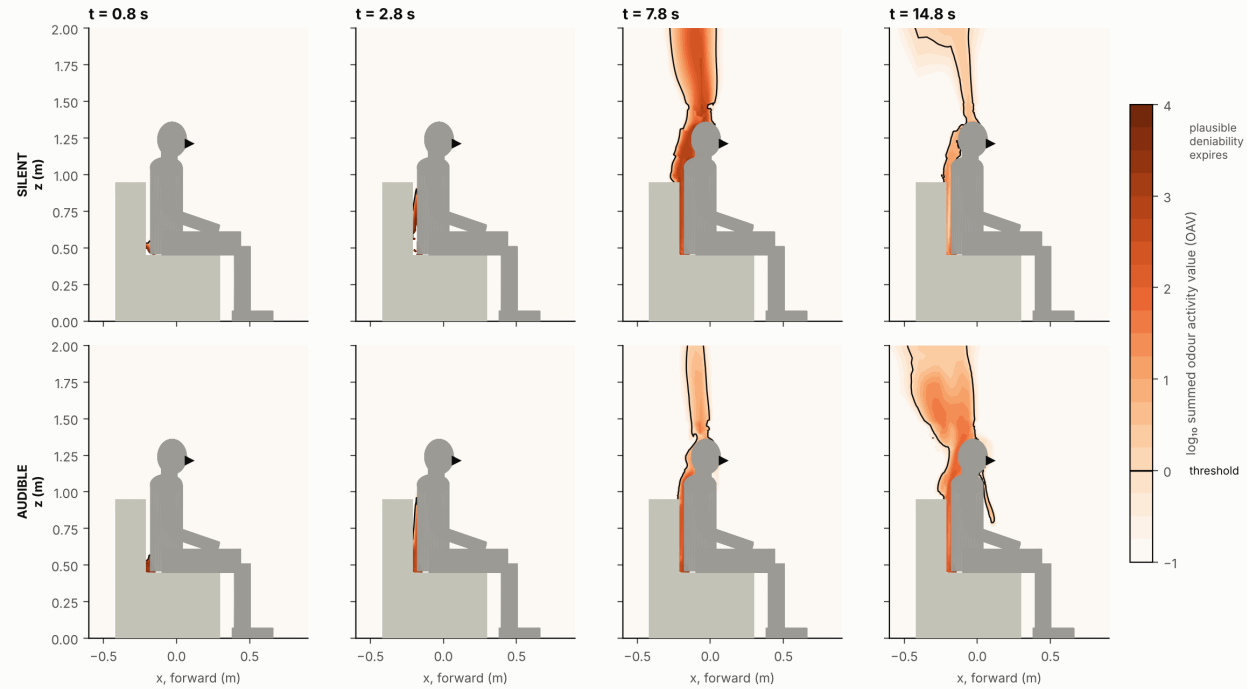


Figure 4: Sofa, releaser's mid-sagittal plane (upright): $\log_{10}$ OAV at 1, 3, 8 and 15 s after the release starts, silent (top) and audible (bottom). The black contour is the odour threshold.

4.5 Perception

Breath sampling. For a neighbour who is not warned, the worst breath still captures 81–99% of the peak. A listener warned by the sound needs about 25 s of breath-hold for an even chance of missing the cloud (Figure S6).

Silent versus audible. Through Stevens' law, the median physical difference at the neighbour becomes a 1% difference in perceived intensity, below the olfactory just-noticeable difference of about 7% in intensity.

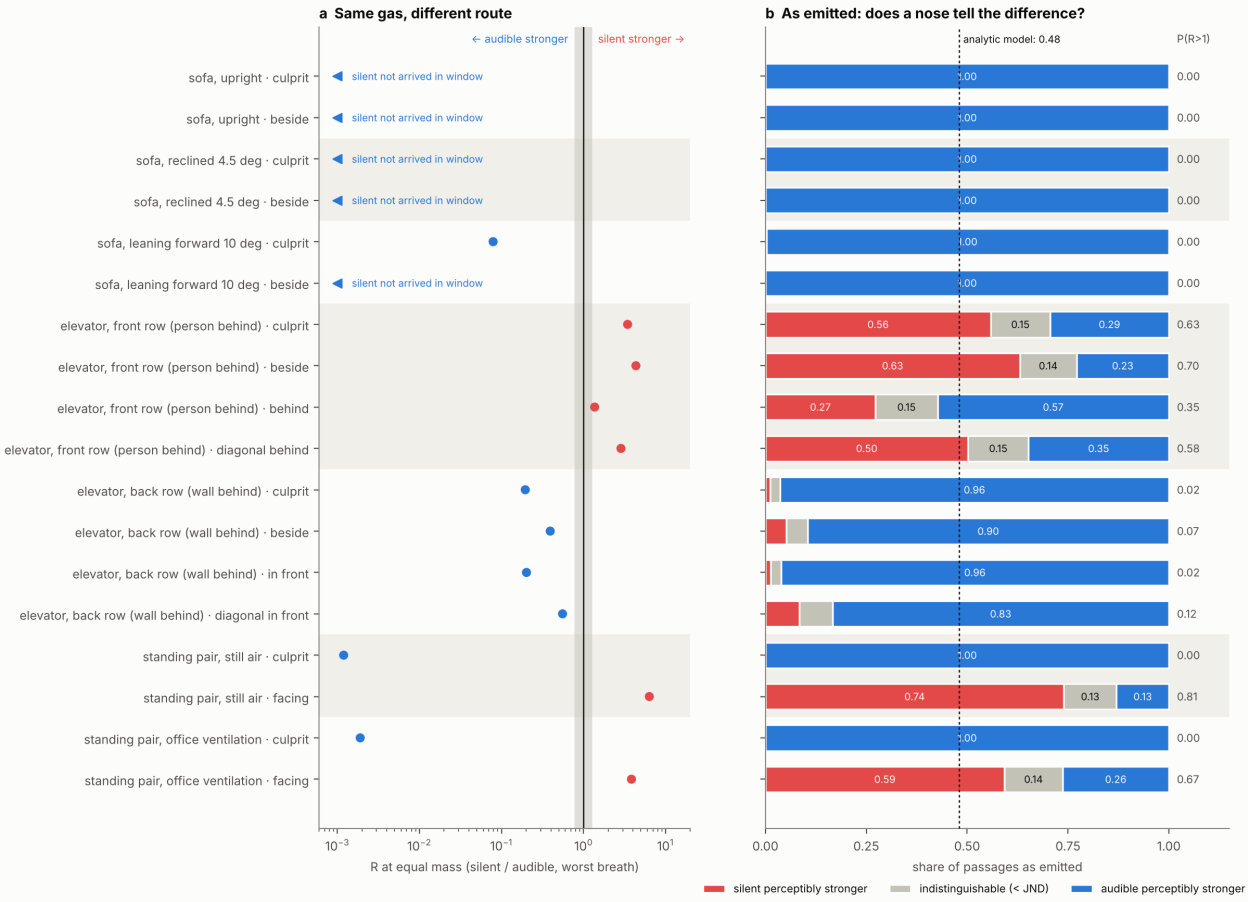


Figure 5: Silent versus audible in the simulated scenes. (a) Ratio of the expected worst-breath OAV (the most concentrated single inhalation) at each nose for the same 92 mL: sofa (30 s, three postures) and elevator (60 s, front-row and back-row releaser). Triangles: the silent release has not reached the nose within the window. The grey band is the just-noticeable ratio. (b) The same flow paths with passages as emitted (20 000 silent/audible volume pairs): shares of pairs in which the silent passage is perceptibly stronger, indistinguishable, or perceptibly weaker, and $P(R > 1)$. The dotted line is the analytic model.

Expectation. An expectation effect would have to exceed 1% in intensity to change which release seems stronger. Measured label effects for brief exposures are 0–3%, and their sign for an unwarned listener has not been established.[63–65] We do not include expectation in the model.

4.6 Memory

Applying Theorem 3 to the simulated passages gives Table 3 and Figure 6. Each base rate p_S assumes the cloud is centred on the neighbour's nose, so it is an upper bound.

Table 3: Share of passages smelled, p_S, and the remembered belief P(a silent passage seems stronger), for literature observer models; "truth" is the same probability over all passages.

observer	p_S	belief	truth
trained panel	0.93	0.53	0.49
attentive, population thresholds	0.59–0.68	0.68–0.73	0.50–0.51
distracted (everyday)	0.21	0.92	0.51
distracted, clothed releaser	0.19	0.93	0.51
distracted, sound cues attention	0.27	0.81	0.32

The fewer passages are smelled, the further the remembered belief departs from the truth. When a sound makes the bystander attentive, audible passages are smelled more often and the belief falls to 0.81; the truth for that observer is also lower (0.32), because they do smell audible passages more often. The releaser, who notices every release, shows no selection (Figure 6a). In loud backgrounds only the largest passages are heard, and this alone weakens the belief, so masking does not isolate the selection mechanism.

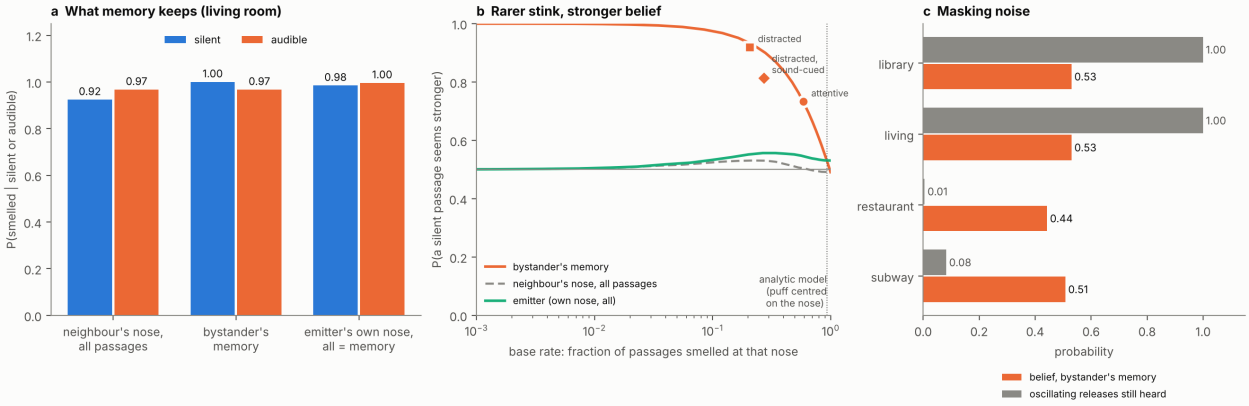


Figure 6: Memory, for 20 000 simulated passages. (a) Probability that a silent or audible passage was smelled: at the neighbour's nose over all passages, in a bystander's memory, and at the releaser's own nose. (b) Probability that a remembered silent passage seems stronger than a remembered audible one, against the share of passages smelled: bystander memory (orange), all passages (grey, dashed) and the releaser (green). Markers show the observer models of Table 3; the dotted line is the analytic model's nominal share. (c) Belief in four backgrounds, with the fraction of oscillating releases still heard.

5 Discussion

5.1 Why the belief exists

The physics we model does not make silent passages generally worse. At equal mass the silent advantage at a neighbour is a median factor of 1.45, often below the just-noticeable ratio. As emitted, a silent passage is the stronger one in 48% of pairs in the analytic model and in [**GAP: v4; preprint: 21% (0–70%)**] of bystander pairs in the simulated scenes, where on a sofa it misses both noses on its first pass.

The belief is nevertheless documented across languages. Theorem 3 can explain it without any physical difference: a passage enters memory if it is heard or smelled, so every remembered silent passage was smelled, whereas audible passages are remembered whether or not they smelled. With literature observer models this selection turns a truth of [**GAP: v4; preprint: 0.32–0.51**] into a remembered belief of [**GAP: v4; preprint: 0.68–0.93**] (Table 3). This shows that selection is sufficient for the belief, not that it is its cause; the predictions below can decide. Three documented memory biases act in the same direction:

distinctive pairings are over-remembered,[66] vivid episodes are easy to recall,[67] and judgements built from biased samples are rarely corrected.[68,69] The phrase itself can also pass the belief on without any experience;[70] documentary evidence cannot separate the two routes (Figure S7).

5.2 Predictions

If selection is the cause, the belief should follow who notices what, and a survey can test this.

- *Releasers.* A releaser feels every release, so their noticing does not depend on smell and Theorem 3 does not apply to their own passages. Releasers should hold the belief less firmly than bystanders.

- *People without a sense of smell.* As bystanders they can notice a passage only by hearing it, so they never remember a silent passage. If they hold the belief, it came from the phrase, not from experience.

- *Background noise.* Masking makes the quieter audible passages seem silent, so they enter memory only when smelled, but the passages still heard are the largest ones (Section 4.6). The two act in opposite directions, so the effect of noise on the belief should be ambiguous and is not a clean test.

5.3 The releaser

Three findings bear on how releasers judge their own passages.

- People rate their own flatus as less disgusting than a stranger's: on a 1–5 scale the item "silent but smells strongly" is rated 2.51 for one's own flatus and 3.71 for a stranger's.[71,72] This lowers the unpleasantness of the releaser's own passages, not the contrast between silent and audible ones.

- The self-advantage appears even when people do not recognise their own odour.[73]
- Believing an odour is one's own shifts its hedonic rating,[74] so expectation is the one route by which a releaser could share the belief.

5.4 Where the odour comes from

Two facts the transport model takes as given have chemical explanations in the literature; we did not test them.

Why sulfur concentration does not follow volume.

- H_2S comes from sulfate-reducing bacteria and from cysteine degradation, whose genes are widespread in the gut microbiome.[75,76] Sulfate reducers and methanogens both consume H_2, but whether they exclude each other is disputed. Sulfidogenic gut contents produced no methane in some studies,[75,77] whereas faecal sulfide did not differ between methane producers and non-producers in others,[78–80] and methanogenic faeces suppressed sulfide production sixfold in co-incubation.[81]
- The caecal mucosa of rats removes both gases quickly: methanethiol is demethylated to H_2S, and H_2S is oxidised mainly to thiosulfate.[82] Rhodanese is the main detoxifying enzyme in human colon,[83] and SELENBP1 oxidises methanethiol with an apparent K_m of 4.8 nM.[84]
- Only free sulfide can enter the gas, and it is less than 5% of faecal total sulfide.[4]

The odorants are thus trace products held at a balance of production and fast consumption, whereas passage volume is set by N_2, CO_2, H_2 and CH_4. We infer that this is why sulfur concentration and passage volume are uncorrelated.[5] Diet moves the balance: a meat-rich diet raised faecal sulfide from 0.22 to 3.38 mmol kg^{-1} in five volunteers,[85] although faecal sulfide reflects the binding capacity of the stool as much as sulfide production,[86] and the response to animal and plant diets differs between people.[87]

Diet can make large passages less sulfurous. Within one diet, sulfur concentration does not change with passage volume (Section 4.1); across diets it can. Rapidly fermented fibres raised gas production up to 50-fold in faecal incubations while suppressing H_2S by up to 89%, and cysteine raised H_2S 16-fold.[88] Dietary sulfate tripled sulfide production and cut methane production fivefold, at unchanged stool weight and transit, in three of six subjects;[89] in sulfidogenic gut contents H_2S replaced CH_4 almost mole for mole,[77] diverting hydrogen from gas volume to sulfide.[90] Interventions aimed at sulfur move sulfur alone: bismuth cut faecal H_2S release 23-fold with total gas unchanged,[91] and in dogs charcoal, yucca and zinc acetate cut H_2S by 38–86% without changing gas production or the number of emissions.[92] Because audible passages are larger, a diet-driven link of this kind would make audible passages less sulfurous per millilitre, in the direction of the belief. Its size in vivo is unknown: no study has measured sulfur gases and flatus volume together across diets.

Why methanethiol smells foul. The human receptor OR2T11 responds to short thiols, including methanethiol, in a copper-dependent way, but not to H_2S.[93] Human nasal mucus contains about 2.5 µM copper,[94] below the 30 µM added in the receptor assays, so the receptor's role in vivo is plausible but not shown. Other receptors respond to sulfur ligands without copper.[95–97] A receptor that responds to methanethiol and not to H_2S would make methanethiol the characteristic note of flatus independently of thresholds; we infer this, it has not been tested.

5.5 When a silent release is worse

At equal mass, at the releaser's own nose, in still air and within a few seconds, a silent release is perceptibly stronger than a loud one (Figure 3a). In the elevator, a silent release is also stronger for the person standing behind a front-row releaser, who smells it first. In these two cases the saying holds.

5.6 Limitations

- *Mean flow only.* The CFD resolves the mean flow; instantaneous peaks of a meandering cloud can exceed the mean by an order of magnitude. This can lift the sub-threshold audible doses on the sofa (OAV 0.2–0.6) above threshold, but not the silent ones (10^{-10}). In the forward-leaning posture the mean silent cloud passes within 25 cm of the neighbour's face, where a filament could reach the nose.

- *Body plume.* The seated body plume is 40–56% too fast, which shortens the travel times of the silent release by about 30% in the seated case; the arrival times in the elevator (8.8 and 17.6 s) are likely short by a similar amount. Standing plumes were not validated separately.

- *Scenes and weights.* The as-emitted shares of Figure 5 weight the simulated scenes equally and assume the flow path does not depend on the passage volume. They describe these scenes, not how often each arises.

- *Base rate.* The share of passages smelled has never been measured; the remembered belief of 0.68–0.93 spans the observer models that bracket it.

- *Valve state.* Manometry shows flatus passing through a relaxed sphincter at a rectal pressure rise of 0–15 mmHg[24,28] but also against a contracted one.[98] The 85% silent share treats the onset pressure as a property of the tissue and does not distinguish the two.

6 Conclusion

- *Routing.* The body's convective boundary layer takes a slow release and lets a fast one pass, so the momentum of a release next to a body decides which nose it reaches. On a sofa the silent release rises behind the releaser to the ceiling and misses both noses on

its first pass; in an elevator it is strongest for the person standing behind the releaser. [**GAP: v4: standing pair**]

- *Peak bound.* Under linear transport a slower release of the same gas never gives a higher peak, in any room or flow. Only the near-field pre-dilution of a fast jet lets a silent release win: by a median factor of 1.45 at equal mass in the analytic model, often below the just-noticeable ratio. As emitted, a silent passage is the stronger one at the neighbour in 48% of pairs in the analytic model and at a bystander in [**GAP: v4; preprint: 21% (0–70%)**] of pairs in the simulated scenes.

- *Selection.* A silent passage is noticed only if it is smelled, and this selection alone can explain the belief: in literature observer models a remembered silent passage seems the stronger one in [**GAP: v4; preprint: 68–93%**] of comparisons. The explanation predicts that releasers hold the belief less firmly than bystanders and that people without a sense of smell never remember a silent passage.

Acknowledgement

[**GAP: AI-use disclosure, required by arXiv and ACS policy: Claude Code (Anthropic) wrote code, ran simulations, searched and extracted literature and drafted text; the author reviewed all content and takes responsibility. To be written last, per the author.**]

Supporting Information Available

Overview diagram; CFD validation; the CFD scenes in detail (sofa postures, elevator, standing pair); breath sampling; the cross-cultural survey; source files of every result; and the full parameter table (PDF). Code and data: [**GAP: Zenodo DOI**].

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